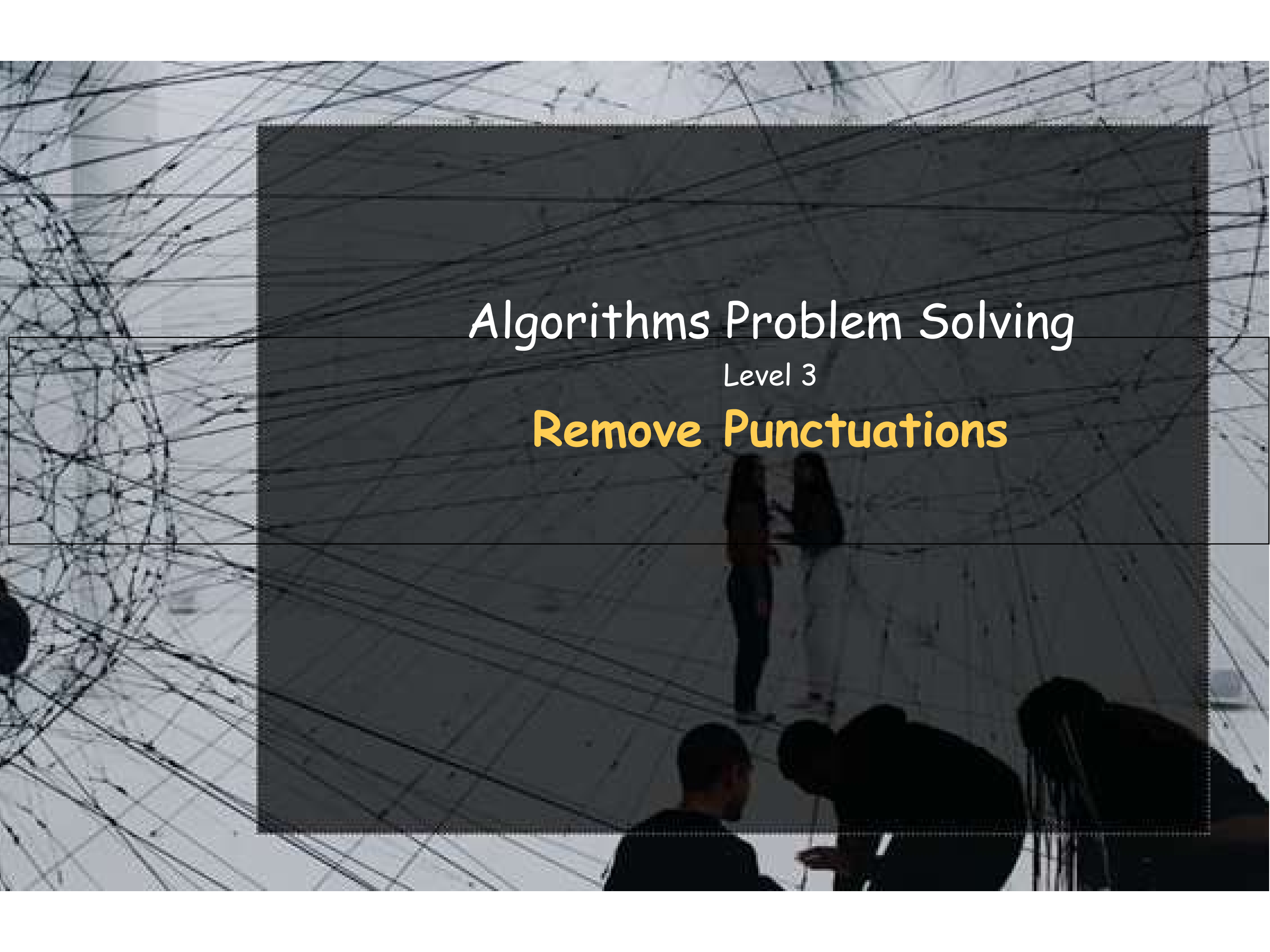


Algorithms Problem Solving

Level 3

Remove Punctuations



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Remove Punctuations

Definition

Punctuation Filtering

-  Program removes punctuation characters from a text string.

-  It preserves letters, digits, and whitespace while filtering characters classified as punctuation.

Explanation

Character Classification

-  Each character is tested using `std::ispunct()` from `<cctype>`.

-  Punctuation characters return a non-zero result from the classification function.

Filtering Logic

-  If a character is not punctuation, it is copied into the result string.

-  Punctuation characters are simply skipped.

Result Construction

-  A new string is built from the characters that pass the filter.

-  The relative order of the remaining characters is preserved.

Procedures

1. Read the complete input string.
2. Create an empty result string.
3. Traverse every character.
4. Check each character with `ispunct()`.
5. Append only non-punctuation characters.
6. Print the cleaned string.

💡 Example

Input:

Welcome to Sweden, it's a nice country!

➡ Result

Welcome to Sweden its a nice country

Note

- ◆ ``ispunct()`` can identify punctuation such as commas, periods, quotation marks, and similar symbols.
- ◆ Spaces are not punctuation, so they remain in the result.
- ◆ Removing punctuation can change the meaning or structure of natural-language text.

⊘ Warning

- ◆ Removing an apostrophe changes `"it's"` into `"its"`.
- ◆ `<cctype>` must be included when using `std::ispunct()`.
- ◆ Character classification should be handled correctly when `char` may be signed.



Comparison

▨ `RemovePunctuationsFromString()`

- ✓ Removes punctuation characters.
- ✓ Preserves letters, digits, and whitespace.
- ✓ Useful for certain normalization tasks.

▤ `RemoveSpacesFromString()`

- ✓ Targets whitespace instead of punctuation.
- ✓ Preserves punctuation marks.
- ✓ Produces a different type of normalized text.

✦ Summary

- ✓ Traverses every character in the input.
- ✓ Uses `std::ispunct()` to identify punctuation.
- ✓ Filters punctuation into a new result string.
- ✓ Preserves the order of all non-punctuation characters.

Thank you 😊